



Review

A meta-analysis of the effects of exposure to microplastics on fish and aquatic invertebrates

Carolyn J. Foley^{a,b,*}, Zachary S. Feiner^a, Timothy D. Malinich^a, Tomas O. Höök^{a,b}

^a Purdue University, Department of Forestry and Natural Resources, 195 Marsteller St., West Lafayette, Indiana 47907, USA

^b Illinois-Indiana Sea Grant College Program, 195 Marsteller St., West Lafayette, Indiana 47907, USA



HIGHLIGHTS

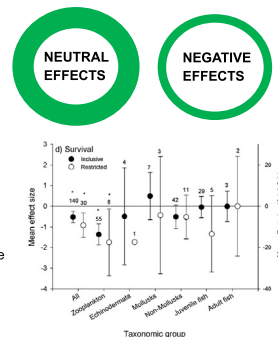
- Microplastics may pose directly deleterious threat to aquatic organisms world-wide.
- Studied impacts of microplastic exposure on aquatic animals for 4 response categories
- Meta-analysis of published studies on consumption, growth, reproduction, survival
- Overall there were few negative impacts, many neutral impacts for these categories.
- Impacts related to microplastic exposure may be more subtle than these categories.

GRAPHICAL ABSTRACT

Effects of exposure to microplastics on fish and aquatic invertebrates



- 4 categories: consumption, growth, reproduction, survival
- 43 published studies; 375 effect sizes
- Summarized findings by taxonomic group and microplastic shape



ARTICLE INFO

Article history:

Received 21 December 2017

Received in revised form 26 February 2018

Accepted 1 March 2018

Available online xxxx

Editor: Daniel Wunderlin

Keywords:

Plastics
Hazard assessment
Microbeads
Microfibers
Review

ABSTRACT

Microplastics are present in aquatic ecosystems the world over and may influence the feeding, growth, reproduction, and survival of freshwater and marine biota; however, the extent and magnitude of potential effects of microplastics on aquatic organisms is poorly understood. In the current study, we conducted a meta-analysis of published literature to examine impacts of exposure to microplastics on consumption (and feeding), growth, reproduction, and survival of fish and aquatic invertebrates. While we did observe within-taxa negative effects for all four categories of responses, many of the effects summarized in our study were neutral, indicating that the effects of exposure to microplastics are highly variable across taxa. The most consistent effect was a reduction in consumption of natural prey when microplastics were present. For some taxa, negative effects on growth, reproduction and even survival were also evident. Organisms that serve as prey to larger predators, e.g., zooplankton, may be particularly susceptible to negative impacts of exposure to microplastic pollution, with potential for ramifications throughout the food web. Future work should focus on whether microplastics may be affecting aquatic organisms more subtly, e.g., by influencing exposure to contaminants and pathogens, or by acting at a molecular level.

© 2018 Elsevier B.V. All rights reserved.

* Corresponding author.

E-mail addresses: cfoley@purdue.edu (C.J. Foley), zfeiner@purdue.edu (Z.S. Feiner), tmalinic@purdue.edu (T.D. Malinich), thook@purdue.edu (T.O. Höök).

Contents

1. Introduction	551
2. Materials and methods	551
3. Results	553
4. Discussion	556
5. Conclusions	557
Acknowledgments	557
References	557

1. Introduction

Plastic pollution is a significant concern for aquatic systems around the world. Microplastics, typically defined as plastic pieces that are <5 mm in size, have been documented in aquatic systems on all seven continents, in both freshwater and marine environments (Barnes et al., 2009), along beaches (Browne et al., 2011), in sediment (Claessens et al., 2011), and in the water column itself (Eriksen et al., 2013). The sources of microplastics available in the wild vary, but include the breakdown of larger plastic items such as food or drink containers, fibers from synthetic clothing, industrial waste, and components of some beauty products (Biginagwa et al., 2016; Kershaw and Rochman, 2015). Though concerns about the effects of plastic pollution on aquatic systems were voiced as early as 1978 (Gregory, 1978; Laist, 1997), the topic has been brought to the forefront in recent years in both the scientific literature and popular media, including such public responses as regulations against the use of microbeads in hygiene products (Pallone, 2015). Researchers and concerned citizens have advocated for information on how microplastics might impact aquatic ecosystems and biota, including fish and aquatic invertebrates (Browne et al., 2007; Seltnerich, 2015). Consequently, the number of studies examining potential impacts of microplastics on aquatic foods webs has increased exponentially (Lusher et al., 2017).

Microplastics have been documented in the digestive systems of wild-caught fish (e.g., Foekema et al., 2013; Phillips and Bonner, 2015) and aquatic invertebrates (e.g., Cole et al., 2011); even in mussels and clams that were ready to be sold for human consumption (Van Cauwenberghe and Janssen, 2014; Davidson and Dudas, 2016). Organisms may directly ingest microplastic particles actively, e.g., due to confusion with potential prey, or passively, e.g., during particle filtration (Collignon et al., 2014). There is some evidence that organisms can avoid consuming microplastics through passive and active selection: *Calanus helgolandicus* and *Acartia clausi* copepods selected prey that were smaller than the microplastics present (Cole et al., 2015; Donaghay and Small, 1979), and *Tripneustes gratilla* sea urchin larvae selected microalgae (food) over polyethylene beads as long as food was present (Kaposi et al., 2014). Conversely, *Eurytemora affinis* copepods ingested latex beads at a higher rate than diatom prey (Powell and Berry, 1990), and holothurian sea cucumbers ingested nylon and polyvinyl chloride fragments rather than sand, potentially because ingesting plastic was easier (Graham and Thompson, 2009). Microplastic can also be incidentally ingested by adhering to natural prey items, e.g. seaweed or fish eggs, (e.g., Kashiwada, 2006; Gutow et al., 2016), or via absorption through gills (e.g., Kashiwada, 2006; Watts et al., 2014). Further, plastic particles that have been ingested could be absorbed through gut walls (Browne et al., 2008; Snell and Hicks, 2011).

Once consumed or otherwise ingested, microplastics can remain in the digestive tracts of aquatic organisms for periods of days to weeks before excretion (e.g., dos Santos and Jobling, 1991; Browne et al., 2008; Cedervall et al., 2012; Batel et al., 2016). This retention time likely allows for the transfer of microplastics both up the food web (e.g., Murray and Cowie, 2011; Farrell and Nelson, 2013), and to new geographic locations (Clark et al., 2016). Exposure of individual aquatic organisms to microplastics may negatively impact feeding (e.g., Wegner et al., 2012;

Ogonowski et al., 2016), growth (e.g., Au et al., 2015; Jeong et al., 2016), reproductive capabilities (e.g., Della Torre et al., 2014; Ogonowski et al., 2016), or survival (e.g., Booth et al., 2016; Luís et al., 2015), due to, for example, blockage of feeding structures or reduced consumption of prey (e.g., as reviewed by Wright et al., 2013b, Eerkes-Medrano et al., 2015). However, the effects of microplastic exposure do not appear to be consistent across studies. Some organisms may be resilient to stresses induced by microplastic exposure (e.g., Nasser and Lynch, 2016; Watts et al., 2016), and the fact that microplastics can be egested suggests cumulative impacts could potentially be minimized. In addition, the shape of plastic particles (e.g., spheres versus fibers) may influence their ability to be ingested and elicit effects on organism performance. Therefore, the overall potential impact of microplastic pollution in aquatic systems remains difficult to predict.

In the current study, we conduct a meta-analysis to examine impacts of exposure to microplastics on four important responses of fish and aquatic invertebrates: a) consumption and feeding (hereafter called “consumption”), b) growth, c) reproduction, and d) survival. Meta-analyses allow researchers to quantitatively assess the effect of a given treatment (in this case, exposure to microplastics) over multiple studies conducted in different locations, on different taxa, and by different research groups to summarize and understand the broader potential impacts of the treatment in question. For each of the four response categories, we use the results reported in published experimental studies to assess whether effects of exposure of fish and aquatic invertebrates to microplastics are positive, negative, or neutral. We further assess whether the effects are consistent across different taxonomic groups or plastic shapes, respectively. Finally, we examine whether the size of the effect varies with experimental conditions including the size of plastic particles used, the temperature at which the experiment was conducted, and the length of time organisms were exposed to microplastics.

2. Materials and methods

We selected studies to include in our meta-analyses via a search of ISI Web of Science on October 14, 2016, using the term “Microplastic* AND growth OR consumption OR fish OR mussel OR zooplankton OR invertebrate”. This initial search yielded 234 papers. Of these, we retained 29 studies for our analyses according to the following criteria: 1) the study examined at least one effect of direct exposure to microplastics on fish or aquatic invertebrate consumption, growth, reproduction, or survival, 2) the study was an experiment (field-based mesocosms were allowed), 3) the study included a “no microplastics” control treatment; and 4) the study reported mean, sample size, and measure of variance for controls and treatments. Forward (i.e., papers that cited each selected paper) and backward (i.e., papers cited by each selected paper) searches performed on this subset of studies in January 2017 yielded an additional 962 papers to examine for additional data. After applying the same criteria for inclusion to these studies, we ultimately included 43 studies in final analyses (Table 1).

We extracted the following information from each study: the species studied; a description of each response included (consumption [e.g., ingestion rate, egestion amount]; growth [e.g., change in weight,

Table 1

The full list of studies included in the meta-analyses plus the number of individual effect sizes included in each of the four “inclusive” datasets. For “Survival”, numbers in brackets reflect effect sizes that could not be converted to odds ratios and were therefore not included in analyses.

Study	Study organism	Number of records included in meta-analyses			
		Consumption	Growth	Reproduction	Survival
Au et al., 2015	<i>Hyalella azteca</i>	3	6	3	8
Avio et al., 2015	<i>Mytilus galloprovincialis</i>				4
Ayukai, 1987	<i>Acartia clausi</i>	6			
Beckingham and Ghosh, 2017	<i>Lumbriculus variegatus</i>		1		
Bergami et al., 2016	<i>Artemia franciscana</i>				10
Besseling et al., 2013	<i>Arenicola marina</i>	8	8		5
Besseling et al., 2014	<i>Daphnia magna</i>		10	10	
Blarer and Burkhardt-Holm, 2016	<i>Gammarus fossarum</i>	2	2		
Booth et al., 2016	<i>Corophium volutator</i>				16
	<i>Daphnia magna</i>				16
Browne et al., 2013	<i>Arenicola marina</i>	6			6
Cole and Galloway, 2015	<i>Crassostrea gigas</i>	8	2		
Cole et al., 2013	<i>Centropages typicus</i>	4			
Cole et al., 2015	<i>Calanus helgolandicus</i>	1		1	1
Della Torre et al., 2014	<i>Paracentrotus lividus</i>			5	
de Sá et al., 2015	<i>Pomatoschistus microps</i>	6			6
Ferreira et al., 2016	<i>Pomatoschistus microps</i>	4			
Fonte et al., 2016	<i>Pomatoschistus microps</i>	10	10		10
Green et al., 2016	<i>Arenicola marina</i>	9	9		
Green, 2016	<i>Ostrea edulis</i>	4	4		
Hämer et al., 2014	<i>Idotea emarginata</i>	7	3		
Imhof and Laforsch, 2016	<i>Potamopyrgus antipodarum</i>		2	4	
Jemec et al., 2016	<i>Daphnia magna</i>		8		16
Jeong et al., 2016	<i>Brachionus koreanus</i>		8	12	(12)
Kaposi et al., 2014	<i>Tripneustes gratilla</i>		4		4
Luis et al., 2015	<i>Pomatoschistus microps</i>	5			6
Ma et al., 2016	<i>Daphnia magna</i>				6
Mazurais et al., 2015	<i>Dicentrarchus labrax</i>		2		2
Nobre et al., 2015	<i>Lytechinus variegatus</i>			2	
Ogonowski et al., 2016	<i>Daphnia magna</i>	2	4	8	(8)
Oliveira et al., 2013	<i>Pomatoschistus microps</i>				5
Pedà et al., 2016	<i>Dicentrarchus labrax</i>		1		1
Rehse et al., 2016	<i>Daphnia magna</i>				6
Rist et al., 2016	<i>Perna viridis</i>	3	3		3
Rochman et al., 2013	<i>Orzyias latipes</i>				2
Rummel et al., 2016	<i>Oncorhynchus mykiss</i>		1		
Sussarellu et al., 2016	<i>Crassostrea gigas</i>	1	1	1	
Tosetto et al., 2016	<i>Platorchestia smithi</i>				2
Ugolini et al., 2013	<i>Talitrus saltator</i>				1
Van Cauwenberghe et al., 2015	<i>Arenicola marina</i>				1
Watts et al., 2016	<i>Carcinus maenas</i>				1
Wegner et al., 2012	<i>Mytilus edulis</i>	3			
Welden and Cowie, 2016	<i>Nephrops norvegicus</i>	1	1		1
Wright et al., 2013a	<i>Arenicola marina</i>	1	3		

length]; reproduction [e.g., fecundity, percentage of abnormal offspring]; survival [e.g., number alive at end of experiment, number of days alive]; Table S1); the shape, type, and mean or minimum size of microplastic used; the temperature at which the study was conducted; the number of days that the organisms were exposed to microplastics; and the sample size, mean, and variation (including type of variation reported) of the response for both control (i.e., no microplastic exposure) and treatment(s). Mean and variation data were extracted from tables when possible. If the data were presented in figures only, we used ImageJ analysis software to physically measure the values (Schneider et al., 2012). A full list of information extracted is included in Table S1.

When there were multiple treatments examined for a particular study and species, e.g., a researcher assessed the response over more than one concentration or type of microplastic, we included an individual record for each treatment that was compared to a (sometimes common) control. Many studies assessed more than one response (e.g., both survival and growth; Table 1). If the authors described multiple experiments for a given response that were independent of each other, e.g., conducted over two discrete time periods, or if the authors reported results of a pilot study as well as the main study, we included records from each experiment. However, if authors described multiple

responses that fell within the same broad category, e.g., they measured growth in terms of both weight and length of organisms, we included only one growth measure in our final dataset. We randomly selected the particular measure to retain for a given response category, after ensuring that the response was easily interpretable in terms of direction and meaning. For studies that reported repeated measurements on the same individual or group of individuals, we included the response at the final time period examined. When survival was measured, we included all responses either from the first instance of 100% mortality in a particular treatment or at the final time period examined.

For consumption, growth, and reproduction-related responses, we calculated effect sizes for each individual record using Hedges' *g* (Hedges and Olkin, 1985), which is essentially Cohen's *d* multiplied by a correction factor (Borenstein et al., 2009).

$$g = J \times (\text{Mean}_T - \text{Mean}_C) / \sqrt{\frac{(n_T - 1) \times SD_T^2 + (n_C - 1) \times SD_C^2}{n_T + n_C - 2}}$$

where *T* and *C* are the values for the treatment and control, respectively, *n* represents their respective sample sizes, and *SD* represents their

respective standard deviations. The correction factor J accounts for small sample sizes and is equal to:

$$J = 1 - \frac{3}{4 \times (n_T + n_C - 9)}$$

Given that Hedges' g requires a pooled standard deviation estimate, if a study reported standard error (SE), we converted it to standard deviation (SD) using $SD = SE \times \sqrt{n}$. If a study reported 95% confidence intervals (CI), we assumed a normal distribution and converted to SD using $SD = CI \times \sqrt{n}/1.96$ (converting reported results to one-way 95% CIs if necessary). If a study reported a range, we estimated SD by dividing the range by 4 (after the Range Rule of Thumb). When accounting for the direction of the effect, we assumed that a higher value was preferred (e.g., higher growth, higher reproductive output, and higher consumption were indicative of improved organism performance). All reproduction studies reported responses that followed this pattern. For the six consumption and eight growth responses that did not follow this pattern, we subtracted the treatment mean from the control mean when calculating Hedges' g . We calculated variance in effect size for each response as

$$V = \frac{(n_C + n_T)}{n_C n_T} + \frac{g^2}{2 \times (n_C + n_T)}$$

Studies that reported effects on survival included a mix of binary, i.e., dead/not dead, and non-binary, i.e., number of days surviving, responses. Wherever possible, for effect sizes in the survival category we calculated the natural logarithm of the odds ratio and variance according to Gurevitch et al. (2001)

$$\ln(OR) = \ln\left(\frac{n_{Treatment,Alive} \times n_{Control,Dead}}{n_{Treatment,Dead} \times n_{Control,Alive}}\right)$$

$$V_{\ln(OR)} = \frac{1}{n_{Treatment,Alive}} + \frac{1}{n_{Treatment,Dead}} + \frac{1}{n_{Control,Alive}} + \frac{1}{n_{Control,Dead}}$$

A positive $\ln(\text{OddsRatio})$ would indicate greater survival for animals exposed to microplastics than control animals; a negative result would indicate lower survival for animals exposed to microplastics than control animals; and a result around 0 would indicate no difference between control and exposure to microplastic treatments. For effect sizes that included a 0 value in one of the four outcome groups (i.e., control + alive, treatment + alive, control + dead, treatment + dead), we added 0.5 to each group (Fleiss, 1981).

For all four responses (consumption, growth, reproduction, survival), we determined the mean effect size according to

$$\text{Mean effect} = \sum E_i W_i / \sum W_i$$

where E_i is the effect size calculated using Hedges' g , or $\ln(\text{odds ratio})$, and W_i is $1/V_i$. To evaluate general microplastic effects, we summarized mean effect sizes plus 95% CI across all studies for each response. To determine whether effects varied by taxa or microplastic shape, we also summarized mean effect sizes and 95% CI by six taxonomic categories (where applicable, all of the following: zooplankton, Echinodermata, mollusk, non-mollusk macroinvertebrate, larval or juvenile fish, and adult fish, calculated across microplastic shapes) and by two general microplastic shapes (rounded, and fiber or fragment, calculated across all taxa). A significant effect of microplastic was determined when the 95% CI did not overlap zero. The vast majority of studies used a round plastic shape (88% of consumption studies; 67% of growth, 83% of reproduction; 86% of survival). Given this highly unbalanced design, we did not separately examine plastic shape effects by taxonomic groups.

Some characteristics of meta-analyses may introduce biases. First, publication bias (i.e., non-reporting of non-significant results) can

potentially skew meta-analyses toward significant outcomes. Second, if multiple effect sizes are derived from a single paper, these measurements may not be truly independent. We examined the potential for publication bias using funnel plots of inverse variance versus effect size for each of the four responses independently, using Egger's tests to assess asymmetry (R package "metafor"; Duval and Tweedie, 2000, Viechtbauer, 2010). Egger's test results using all effect sizes extracted (hereafter referred to as the "inclusive" dataset) suggested asymmetry for consumption ($z = -5.4$, $p < 0.0001$), growth ($z = -8.4$, $p < 0.0001$), reproduction ($z = -6.8$, $p < 0.0001$), and survival ($z = -2.1$, $p = 0.03$) (Fig. 1). We attempted to account for any bias introduced by our decision to include multiple effect sizes from a single paper by repeating the above analyses using four "restricted" datasets (one each for consumption, growth, reproduction, and survival). To create these four datasets, we first removed obvious outliers via visual inspection (Fig. 1). We then selected the presumed most extreme treatment from each paper (i.e., highest concentration of microplastics, longest exposure time, highest temperature, color of plastic most likely to represent prey) of those effect sizes that remained. In some cases, we randomly selected a treatment effect (e.g., if an experiment exposed two different strains of fish). The full list of effect sizes included in the restricted dataset is noted in Table S1. Egger's tests performed on the restricted datasets (Fig. 2) still indicated asymmetry for growth ($z = -2.4$, $p = 0.019$), and reproduction ($z = -3.2$, $p = 0.0013$), but not for consumption ($z = -1.8$, $p = 0.08$), or survival ($z = -0.59$, $p = 0.55$).

To examine which factors might exacerbate effects of exposure to microplastics, we conducted linear regressions between effect size and log-transformed mean or minimum size of microplastic used; temperature at which the study was conducted; and number of days that the organisms were exposed to microplastics, respectively, in R (R Core Team, 2016), using the "inclusive" dataset only (i.e., all effect sizes extracted).

3. Results

We discuss our findings in terms of both the "inclusive" and "restricted" datasets, and generally suggest that the strength of a particular effect size pattern (and any subsequent conclusions drawn) relates to whether or not it holds across both analyses. Microplastic ingestion was confirmed in 81% of the studies included ($n = 43$). Studies tested a variety of microplastic exposure conditions, including exposing animals to multiple types of plastics, holding animals at different temperatures, exposing different tanks to different types or concentrations of microplastics, exposing to microplastics in the presence of other contaminants, and exposing over different time periods.

For the inclusive datasets, we included 94 individual effect size calculations from 21 different studies in the consumption meta-analysis, 93 individual effect size calculations from 22 different studies in the growth meta-analysis, 46 individual effect size calculations from 9 different studies in the reproduction meta-analysis, and 139 individual effect size calculations from 24 different studies in the survival meta-analysis (Table 1). For the restricted datasets, we included 27 individual effect size calculations from 19 different studies in the consumption meta-analysis, 31 individual effect size calculations from 22 different studies in the growth meta-analysis, 10 individual effect size calculations from 9 different studies in the reproduction meta-analysis, and 30 individual effect size calculations from 24 different studies in the survival meta-analysis.

The effect of exposure to microplastics varied with taxonomic group (Fig. 3). Summarizing across all groups, exposure to microplastics had a significantly negative effect on consumption, growth, reproduction, and survival, even when using restricted datasets (except for reproduction; Fig. 3). Zooplankton growth, reproduction, and survival were all significantly reduced when exposed to microplastics. In general, our results do not demonstrate that Echinodermata and mollusks are affected by exposure to microplastics. There is, however, some suggestion that echinoderm reproduction, and mollusk consumption and growth, may be

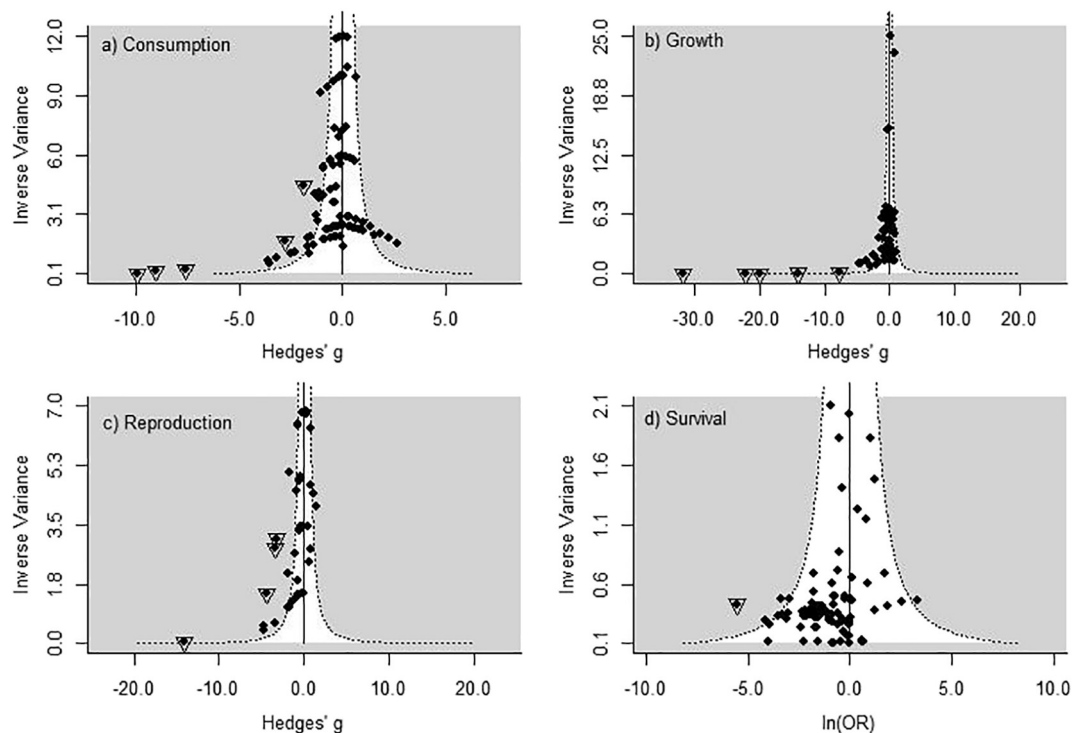


Fig. 1. Funnel plots for the “inclusive” dataset used to check potential for publication bias. Dotted lines depict the expected shape of the funnel based on pseudo-confidence calculations. Outliers depicted with a triangle were not considered for inclusion in the “restricted” dataset (see Methods for details).

negatively affected (Fig. 3). Non-mollusk macroinvertebrate growth was marginally negatively affected (Fig. 3). Larval and juvenile fish consumption and feeding was significantly negatively affected by exposure to microplastics. While our results do not demonstrate that adult fish were affected, the number of studies that used adult fish as test organisms was quite low (Fig. 3).

In general, exposure to round plastic particles was associated with negative effects on all four responses (Fig. 4). Exposure to fibrous or fragmented plastic particles also negatively affected survival, but effects on other responses varied from negative to neutral (Fig. 4).

Using the inclusive dataset, we observed only one strong, significant relationship between a given effect size and the log-transformed

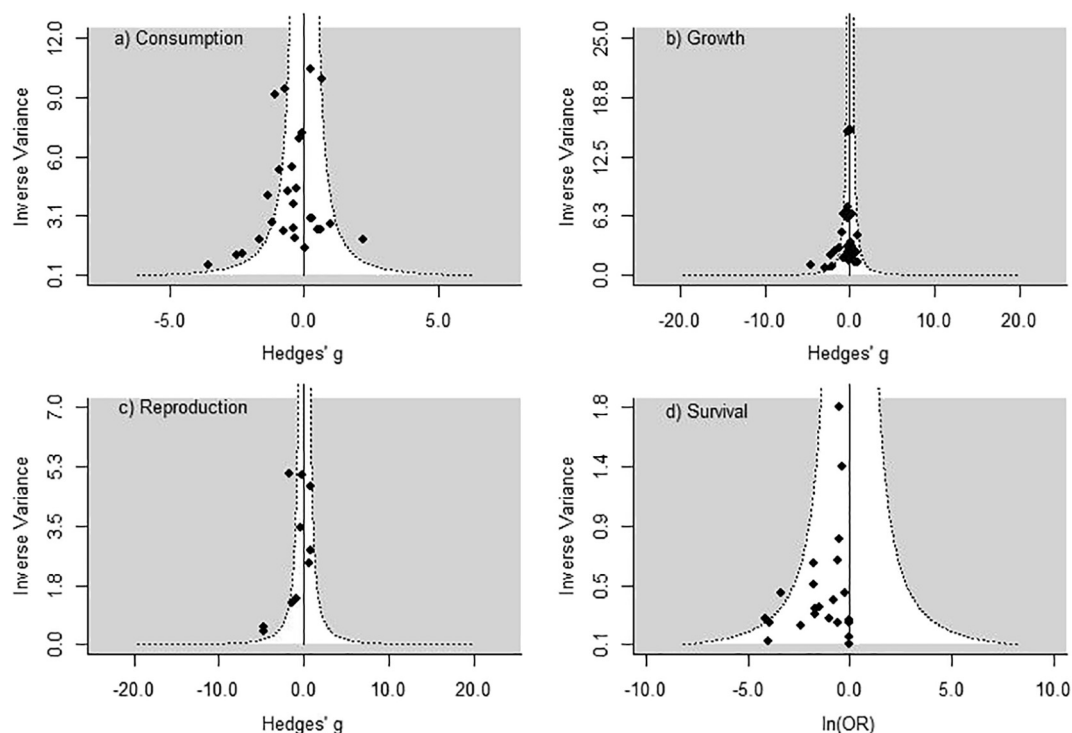


Fig. 2. Funnel plots for the “restricted” dataset used to check potential for publication bias. Dotted lines depict the expected shape of the funnel based on pseudo-confidence calculations.

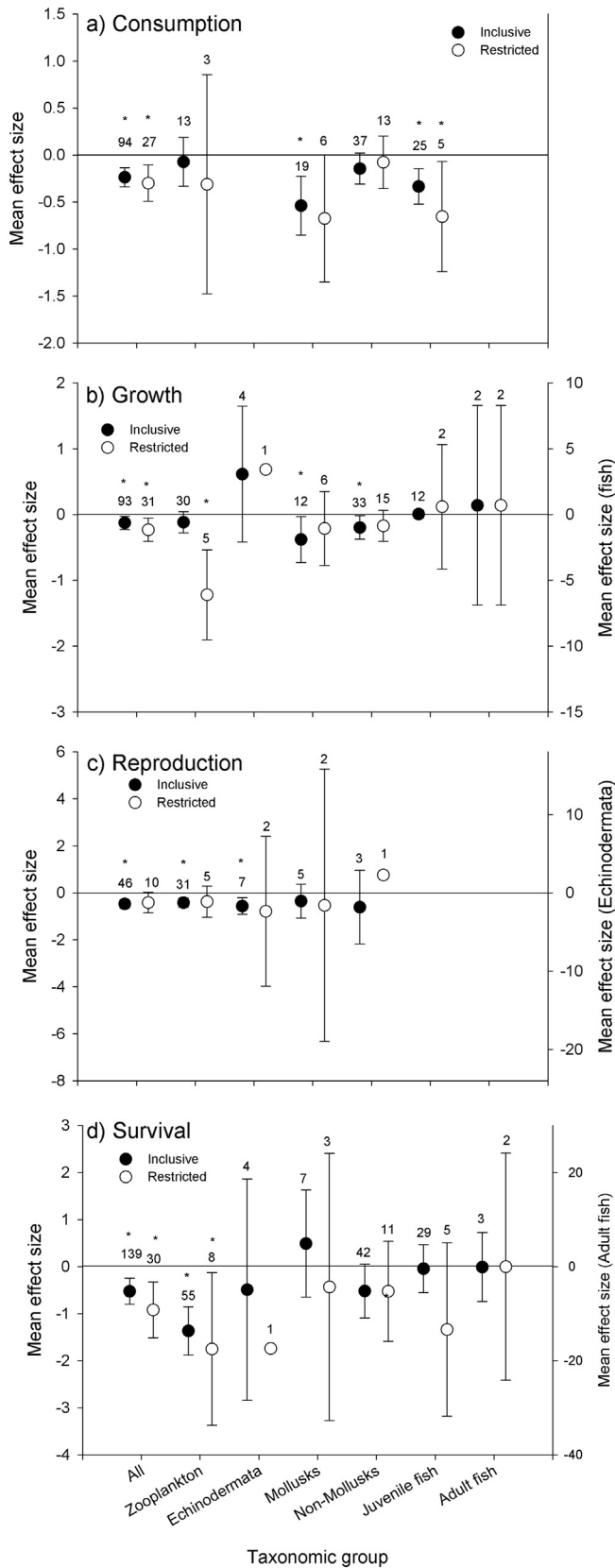


Fig. 3. Mean effect sizes $\pm$ 95% CI for the four different response categories (a–d) broken down by taxonomic group. Closed circles indicate the “inclusive” dataset while open circles indicate the “restricted” dataset. Effect sizes calculated as Hedges’ g for (a–c), and $\ln(\text{odds ratio})$ for (d). Sample sizes noted above each bar. “Juvenile” fish may include juvenile or larval fish. Significant effects are marked with *. Note different scale for fish in (b) and (d) and Echinodermata in (c).

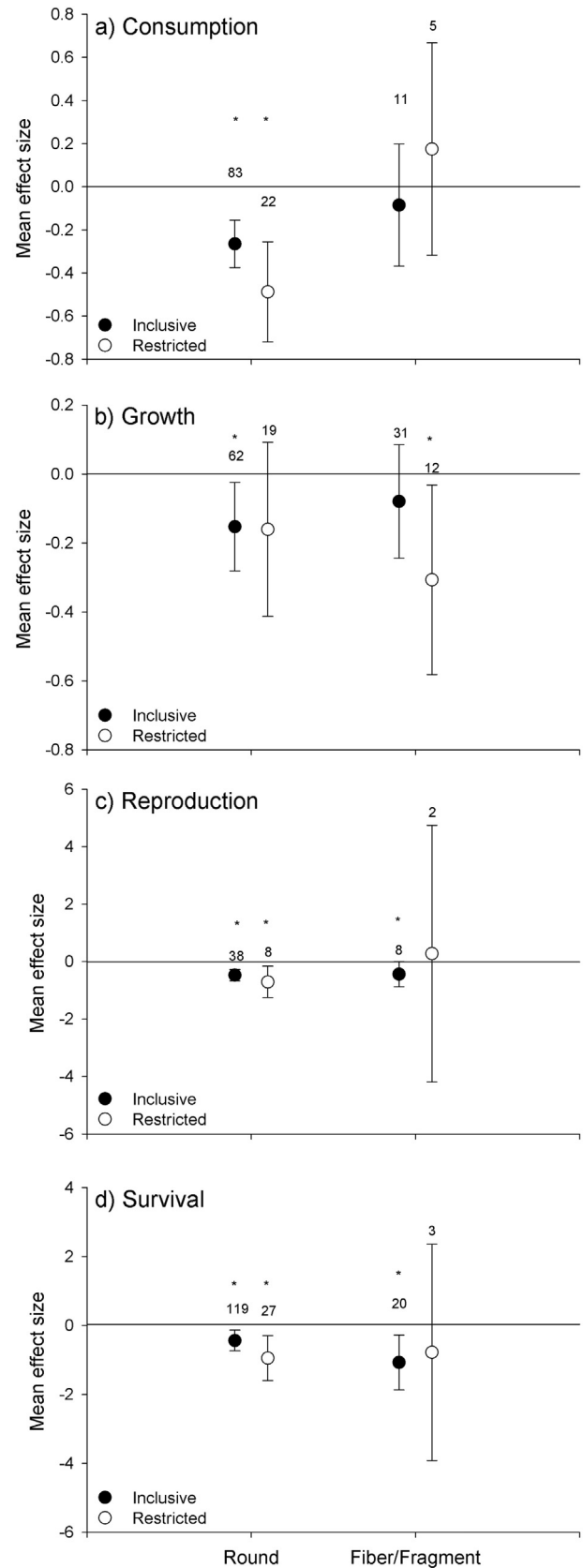


Fig. 4. Mean effect sizes $\pm$ 95% CI for the four different response categories (a–d) broken down by generalized microplastic shape. Closed circles indicate the “inclusive” dataset while open circles indicate the “restricted” dataset. Effect sizes calculated as Hedges’ g for (a–c), and $\ln(\text{odds ratio})$ for (d). Sample sizes noted above each bar. Significant effects are marked with *.

Table 2Results of regression analyses between effect size (dependent variable = Hedges' g or $\ln(\text{odds ratio})$) and three independent variables extracted from studies.

Response category	Independent variable (log-transformed)	β	F	df	R ²	p-Value
Consumption	Min. or mean size of microplastic (μm)	−0.11	0.91	1, 92	0.01	0.34
	Temperature ($^{\circ}\text{C}$)	−1.73	1.48	1, 79	0.018	0.23
	Max. exposure to microplastics (days)	0.29	6.67	1, 92	0.068	0.01
Growth	Min. or mean size of microplastic (μm)	0.098	0.62	1, 90	0.0068	0.43
	Temperature ($^{\circ}\text{C}$)	0.18	0.016	1, 76	0.0002	0.9
	Max. exposure to microplastics (days)	−0.84	22.2	1, 91	0.2	<0.0001
Reproduction	Min. or mean size of microplastic (μm)	0.24	0.98	1, 42	0.023	0.33
	Temperature ($^{\circ}\text{C}$)	−0.43	0.016	1, 42	0.00039	0.9
	Max. exposure to microplastics (days)	−0.10	0.056	1, 44	0.0018	30.81
Survival	Min. or mean size of microplastic (μm)	0.01	0.43	1, 136	0.0032	0.51
	Temperature ($^{\circ}\text{C}$)	0.34	1.35	1, 129	0.01	0.25
	Max. exposure to microplastics (days)	0.0065	0.067	1, 137	0.0005	0.8

temperature, log-transformed mean or minimum size of plastic in experiment, or log-transformed number of days exposed to plastic, respectively (Table 2). Specifically, growth effect sizes were significantly negatively related to number of days exposed to microplastics; however, this relationship was driven by one outlier (Table S1).

4. Discussion

A number of scientific studies assessing the impacts of microplastics on the vital rates of fish and aquatic invertebrates have been reviewed (e.g., Eerkes-Medrano et al., 2015; Phuong et al., 2016; Wright et al., 2013b, among others), but have heretofore not been systematically analyzed for general, taxa-, or shape-specific trends in microplastic effects. Our findings suggest that many aquatic biota have the potential to be negatively impacted by exposure to microplastics; however, there were few generalizable trends consistent across taxonomic grouping or shape of plastic. Our results most strongly support the notion that exposure to microplastics leads to negative effects on consumption of aquatic organisms, with less compelling and consistent evidence that growth, reproduction, or survival of aquatic organisms is negatively affected by exposure to microplastics.

Our results suggest that zooplankton are among the most susceptible biota to microplastic exposure, as their growth, survival, and, when all data were included, reproduction were negatively affected. This has potential for broader ramifications within aquatic food webs, as zooplankton serve as important prey and consumers in aquatic systems across the world (e.g., Sheldon et al., 1977; Sommer et al., 2002). Other animals included in our analyses, e.g. sea urchins, amphipods and isopods, also play a role in structuring marine food webs (e.g., Paine, 1992). If the vital rates of these taxa are negatively impacted by exposure to microplastics, then their populations may decline, with potential effects throughout local food webs (Pace et al., 1999). In addition, as others have noted, the tendency of these taxa to consume microplastics may promote the accumulation and transfer of plastics up the food web (e.g., Setälä et al., 2014; Farrell and Nelson, 2013).

Our findings support the notion that plastics interfere directly with feeding by larval or juvenile fishes, potentially blocking digestive tracts or otherwise not allowing for proper digestive function (reviewed in Cole et al., 2011). Growth and survival of small-bodied aquatic animals is clearly linked to their ability to ingest sufficient calories (e.g., growing large enough to escape gape-limited predators or survive in harsh over winter conditions; Sogard, 1997), therefore any factor that negatively influences an animal's ability to feed may have impacts on long-term growth and survival. Interestingly, our findings do not provide strong evidence that growth was negatively impacted by plastic exposure. It is possible that many studies did not extend long enough for strong growth effects to be observed, given that most exposures were limited to <30 days.

While we did not see strong evidence that size of microplastic, length of exposure to microplastic, or experimental temperature was

related to the effect of microplastic exposure on aquatic fish or invertebrates, it could have been insightful to similarly examine the relationship between effect size and concentration of microplastics animals were exposed to. Presumably, exposure to greater concentrations of microplastic would lead to a stronger effect, but there may be a threshold above which additional microplastic exposure does not correlate with increased risk to an organism. However, methods for reporting microplastic concentrations were highly variable from study to study, precluding a comprehensive analysis of concentration effects. Some authors reported the number of microplastic pieces per unit volume (e.g., Blarer and Burkhardt-Holm, 2016; de Sá et al., 2015), others the weight of microplastic per unit volume (e.g., Booth et al., 2016; Fonte et al., 2016), and still others the percent plastic by weight of sediment or food (e.g., Browne et al., 2013; Hämer et al., 2014). While standardization of how concentrations are reported across all studies may not be feasible (after Rochman et al., 2017), authors should consider reporting both the size and weight of individual plastic particles, if possible, and a weight or density per unit of volume (as described in Phuong et al., 2016).

The four response types we assessed are only part of a wide range of physiological and biochemical responses previously examined, where the broader set included examination of physical malformations (e.g., Kaposi et al., 2014; Besseling et al., 2014; Pedà et al., 2016), and altered endocrine function, metabolite composition, and gene expression (e.g., Rochman et al., 2013; Karami et al., 2016; Lu et al., 2016). As opposed to the relatively direct responses that we assessed, it is possible that effects of exposure to microplastics are more indirect (e.g., alteration of microbial communities in the environment or guts; Oberbeckmann et al., 2015) or have more direct and apparent impacts on responses other than the four assessed herein (e.g., endocrine disruptor effects that negatively impact reproduction; Sussarellu et al., 2016). Examining biochemical data in a way similar to the current study may be challenging, as quantification of these responses is sometimes qualitative (e.g., presence rather than degree of fluorescence) and the specific responses may vary among studies with unclear cross-taxa interpretations (e.g., regulation of specific gene expression; Mazurais et al., 2015). The biochemical effects of microplastics have potentially important implications for the fitness of organisms (e.g., Rochman et al., 2013), and a comprehensive examination of their effects is likely warranted.

A major concern not addressed in the current study is whether contaminants that have adsorbed onto microplastics might be passed to organisms upon ingestion (Koelmans et al., 2016; Wardrop et al., 2016). We attempted to directly compare effect sizes within studies that exposed animals to an additional contaminant besides microplastic (i.e., to isolate the effect of microplastics); however, extremely small sample sizes ($n < 8$) limited our ability to make inferences about contaminant effects. The degree of interaction between microplastics and other contaminants was not predictable a priori. In previous studies, PBDE-spiked plastics likely did not serve as a delivery route of PBDEs

to amphipod tissues (Chua et al., 2014), PCB concentrations in fish tissue decreased after fish were fed PCB-spiked food followed by clean plastic (Rummel et al., 2016), and exposing organisms to silver or fluoranthene alongside microplastics may have helped decrease the amount of contaminant that was ultimately transferred to organisms (Khan et al., 2015; Paul-Pont et al., 2016). All of these authors noted, however, that any decreases in contaminant level could also have been attributed to other sources, and transfer of contaminants to organisms did still happen (Khan et al., 2015; Paul-Pont et al., 2016; Rummel et al., 2016). Given that microplastics are one of many stressors acting on fish and aquatic invertebrates in natural systems, we suggest that studies on interaction effects remain a research priority in the future.

As even the restricted datasets for growth and reproduction demonstrated asymmetry in the funnel plots, the results included in our analyses were potentially affected by publication bias. This remains a challenge to meta-analyses, and even the studies that we included had bias-related issues (e.g., non-significant results were sometimes included as supplementary rather than main body material; some authors simply stated that results were not significant and data were not shown). If we are to fully understand impacts of emerging issues, it is critical that results achieved via sound scientific methods be included in the publication record, regardless of their statistical significance on a case-by-case basis. Another issue with meta-analyses is lack of reporting full sets of information. Several studies included in our analyses did not report the type of error that was presented in tables or figures. In all but one case, we were able to contact authors and confirm the type of error that had been reported. However, we urge researchers to be sure to include these details in their work.

5. Conclusions

The effects of microplastic exposure represents a major issue in nearly every aquatic system on the planet (Barnes et al., 2009), no matter how remote from potential pollution sources (e.g., Free et al., 2014). This paper represents the first systematic meta-analysis of microplastic effects across a range of marine and freshwater taxa. Our findings support the scientific and public concern over plastic pollution of aquatic ecosystems: effects of microplastics were generally negative or neutral across taxa (never positive), with the strongest effects observed on lower trophic level organisms that serve as important linchpins for food web structure (Pace et al., 1999). Beyond our direct results, our analysis also reinforces the need for a standardized method for reporting and evaluating microplastic densities in aquatic habitats and experiments, a better understanding of the biochemical effects of microplastics on organisms, and evaluation of the interaction between microplastics and contaminants that may influence their impacts on organism performance and bioaccumulation of important chemicals, like PBDEs, through aquatic food webs. Understanding these dynamics will provide critical information elucidating the potential food web effects of microplastics in aquatic systems, up to and including their potential ramifications for fisheries productivity and contaminant bioaccumulation that could ultimately affect species regularly consumed by humans (Rochman et al., 2013).

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.scitotenv.2018.03.046>.

Acknowledgments

The authors wish to acknowledge the Purdue University interlibrary loan staff. Animal and microplastic images in the graphical abstract are used courtesy of NOAA-GLERL and Irene Miles of Illinois-Indiana Sea Grant. Two anonymous reviewers improved an earlier version of the manuscript. This work was supported by the Illinois-Indiana Sea Grant College Program (NA14OAR4170095, NA18OAR4170082) and the Department of Forestry and Natural Resources, Purdue University.

References

- Au, S.Y., Bruce, T.F., Bridges, W.C., Klaine, S.J., 2015. Responses of *Hyalella azteca* to acute and chronic microplastic exposures. *Environ. Toxicol. Chem.* 34:2564–2572. <https://doi.org/10.1002/etc.3093>.
- Avio, C.G., Corbi, S., Milan, M., Benedetti, M., Fattorini, D., d'Errico, G., Pauleto, M., Bargelloni, L., Regoli, F., 2015. Pollutants bioavailability and toxicological risk from microplastics to marine mussels. *Environ. Pollut.* 198:211–222. <https://doi.org/10.1016/j.envpol.2014.12.021>.
- Ayukai, T., 1987. Discriminate feeding of the calanoid copepod *Acartia clausi* in mixtures of phytoplankton and inert particles. *Mar. Biol.* 94:579–587. <https://doi.org/10.1007/BF00431404>.
- Barnes, D.K.A., Gargani, F., Thompson, R.C., Barlaz, M., 2009. Accumulation and fragmentation of plastic debris in global environments. *Philos. Trans. R. Soc., B* 364, 1985–1998.
- Batel, A., Linti, F., Scherer, M., Erdinger, L., Braunbeck, T., 2016. Transfer of benzo[a]pyrene from microplastics to *Artemia nauplii* and further to zebrafish via a trophic food web experiment: CYP1A induction and visual tracking of persistent organic pollutants. *Environ. Toxicol. Chem.* 35:1656–1666. <https://doi.org/10.1002/etc.3361>.
- Beckingham, B., Ghosh, U., 2017. Differential bioavailability of polychlorinated biphenyls associated with environmental particles: Microplastic in comparison to wood, coal and biochar. *Environ. Pollut.* 220:150–158. <https://doi.org/10.1016/j.envpol.2016.09.033>.
- Bergami, E., Bocci, E., Vannuccini, M.L., Monopoli, M., Salvati, A., Dawson, K.A., Corsi, I., 2016. Nano-sized polystyrene affects feeding, behavior and physiology of brine shrimp *Artemia franciscana* larvae. *Ecotoxicology and Environmental Safety*, 6th Biannual ECO-toxicology MEeting (BECOME 2014) - Environmental Emergencies: Ecotoxicology as a Management Tool. 123:pp. 18–25. <https://doi.org/10.1016/j.ecoenv.2015.09.021>.
- Besseling, E., Wegner, A., Foekema, E.M., van den Heuvel-Greve, M.J., Koelmans, A.A., 2013. Effects of microplastic on fitness and PCB bioaccumulation by the Lugworm *Arenicola marina* (L.). *Environ. Sci. Technol.* 47:593–600. <https://doi.org/10.1021/es302763x>.
- Besseling, E., Wang, B., Lüring, M., Koelmans, A.A., 2014. Nanoplastic affects growth of *S. obliquus* and reproduction of *D. magna*. *Environ. Sci. Technol.* 48:12336–12343. <https://doi.org/10.1021/es503001d>.
- Biginagwa, F.J., Mayoma, B.S., Shashoua, Y., Syberg, K., Khan, F.R., 2016. First evidence of microplastics in the African Great Lakes: recovery from Lake Victoria Nile perch and Nile tilapia. *J. Great Lakes Res.* 42:146–149. <https://doi.org/10.1016/j.jglr.2015.10.012>.
- Blarer, P., Burkhardt-Holm, P., 2016. Microplastics affect assimilation efficiency in the freshwater amphipod *Gammarus fossarum*. *Environ. Sci. Pollut. Res. Int.* 23:23522–23532. <https://doi.org/10.1007/s11356-016-7584-2>.
- Booth, A.M., Hansen, B.H., Frenzel, M., Johnsen, H., Altin, D., 2016. Uptake and toxicity of methylmethacrylate-based nanoplastic particles in aquatic organisms. *Environ. Toxicol. Chem.* 35:1641–1649. <https://doi.org/10.1002/etc.3076>.
- Borenstein, et al., 2009. In: Borenstein, Michael, Hedges, Larry V., Higgins, Julian P.T., et al. (Eds.), Wiley: Introduction to Meta-Analysis [WWW Document], n.d. URL: <http://www.wiley.com/WileyCDA/WileyTitle/productCd-EHEP002313.html>, Accessed date: 5 December 2017.
- Browne, M.A., Galloway, T., Thompson, R., 2007. Microplastic—an emerging contaminant of potential concern? *Integr. Environ. Assess. Manag.* 3:559–561. <https://doi.org/10.1002/ieam.5630030412>.
- Browne, M.A., Dissanayake, A., Galloway, T.S., Lowe, D.M., Thompson, R.C., 2008. Ingested microscopic plastic translocates to the circulatory system of the mussel, *Mytilus edulis* (L.). *Environ. Sci. Technol.* 42:5026–5031. <https://doi.org/10.1021/es800249a>.
- Browne, M.A., Crump, P., Niven, S.J., Teuten, E., Tonkin, A., Galloway, T., Thompson, R., 2011. Accumulation of microplastic on shorelines worldwide: sources and sinks. *Environ. Sci. Technol.* 45:9175–9179. <https://doi.org/10.1021/es201811s>.
- Browne, M.A., Niven, S.J., Galloway, T.S., Rowland, S.J., Thompson, R.C., 2013. Microplastic moves pollutants and additives to worms, reducing functions linked to health and biodiversity. *Curr. Biol.* 23:2388–2392. <https://doi.org/10.1016/j.cub.2013.10.012>.
- Cedervall, T., Hansson, L.-A., Lard, M., Frohm, B., Linse, S., 2012. Food chain transport of nanoparticles affects behaviour and fat metabolism in fish. *PLoS ONE* 7, e32254. <https://doi.org/10.1371/journal.pone.0032254>.
- Chua, E.M., Shimeta, J., Nuggeoda, D., Morrison, P.D., Clarke, B.O., 2014. Assimilation of polybrominated diphenyl ethers from microplastics by the marine amphipod, *Allochordates compressa*. *Environ. Sci. Technol.* 48:8127–8134. <https://doi.org/10.1021/es405717z>.
- Claessens, M., Meester, S.D., Landuyt, L.V., Clerck, K.D., Janssen, C.R., 2011. Occurrence and distribution of microplastics in marine sediments along the Belgian coast. *Mar. Pollut. Bull.* 62:2199–2204. <https://doi.org/10.1016/j.marpolbul.2011.06.030>.
- Clark, J.R., Cole, M., Lindeque, P.K., Fileman, E., Blackford, J., Lewis, C., Lenton, T.M., Galloway, T.S., 2016. Marine microplastic debris: a targeted plan for understanding and quantifying interactions with marine life. *Front. Ecol. Environ.* 14:317–324. <https://doi.org/10.1002/fee.1297>.
- Cole, M., Galloway, T.S., 2015. Ingestion of nanoplastics and microplastics by Pacific oyster larvae. *Environ. Sci. Technol.* 49:14625–14632. <https://doi.org/10.1021/acs.est.5b04099>.
- Cole, M., Lindeque, P., Halsband, C., Galloway, T.S., 2011. Microplastics as contaminants in the marine environment: a review. *Mar. Pollut. Bull.* 62:2588–2597. <https://doi.org/10.1016/j.marpolbul.2011.09.025>.
- Cole, M., Lindeque, P., Fileman, E., Halsband, C., Goodhead, R., Moger, J., Galloway, T.S., 2013. Microplastic ingestion by zooplankton. *Environ. Sci. Technol.* 47:6646–6655. <https://doi.org/10.1021/es400663f>.
- Cole, M., Lindeque, P., Fileman, E., Halsband, C., Galloway, T.S., 2015. The impact of polystyrene microplastics on feeding, function and fecundity in the marine copepod *Calanus helgolandicus*. *Environ. Sci. Technol.* 49:1130–1137. <https://doi.org/10.1021/es504525u>.

- Collignon, A., Hecq, J.-H., Galgani, F., Collard, F., Goffart, A., 2014. Annual variation in neustonic micro- and meso-plastic particles and zooplankton in the Bay of Calvi (Mediterranean-Corsica). *Mar. Pollut. Bull.* 79:293–298. <https://doi.org/10.1016/j.marpolbul.2013.11.023>.
- Davidson, K., Dudas, S.E., 2016. Microplastic ingestion by wild and cultured Manila clams (*Venerupis philippinarum*) from Baynes Sound, British Columbia. *Arch. Environ. Contam. Toxicol.* 71:147–156. <https://doi.org/10.1007/s00244-016-0286-4>.
- de Sá, L.C., Luís, L.G., Guilhermino, L., 2015. Effects of microplastics on juveniles of the common goby (*Pomatoschistus microps*): confusion with prey, reduction of the predatory performance and efficiency, and possible influence of developmental conditions. *Environ. Pollut.* 196:359–362. <https://doi.org/10.1016/j.envpol.2014.10.026>.
- Della Torre, C., Bergami, E., Salvati, A., Faleri, C., Cirino, P., Dawson, K.A., Corsi, I., 2014. Accumulation and embryotoxicity of polystyrene nanoparticles at early stage of development of sea urchin embryos *Paracentrotus lividus*. *Environ. Sci. Technol.* 48:12302–12311. <https://doi.org/10.1021/es502569w>.
- Donaghay, P.L., Small, L.F., 1979. Food selection capabilities of the estuarine copepod *Acartia clausi*. *Mar. Biol.* 52:137–146. <https://doi.org/10.1007/BF00390421>.
- dos Santos, J., Jobling, M., 1991. Gastric emptying in cod, *Gadus morhua* L.: emptying and retention of indigestible solids. *J. Fish Biol.* 38:187–197. <https://doi.org/10.1111/j.1095-8649.1991.tb03105.x>.
- Duval, S., Tweedie, R., 2000. Trim and fill: a simple funnel-plot-based method of testing and adjusting for publication bias in meta-analysis. *Biometrics* 56, 455–463.
- Eerkes-Medrano, D., Thompson, R.C., Aldridge, D.C., 2015. Microplastics in freshwater systems: a review of the emerging threats, identification of knowledge gaps and prioritisation of research needs. *Water Res.* 75:63–82. <https://doi.org/10.1016/j.watres.2015.02.012>.
- Eriksen, M., Mason, S., Wilson, S., Box, C., Zellers, A., Edwards, W., Farley, H., Amato, S., 2013. Microplastic pollution in the surface waters of the Laurentian Great Lakes. *Mar. Pollut. Bull.* 77:177–182. <https://doi.org/10.1016/j.marpolbul.2013.10.007>.
- Farrell, P., Nelson, K., 2013. Trophic level transfer of microplastic: *Mytilus edulis* (L.) to *Carcinus maenas* (L.). *Environ. Pollut.* 177:1–3. <https://doi.org/10.1016/j.envpol.2013.01.046>.
- Ferreira, P., Fonte, E., Soares, M.E., Carvalho, F., Guilhermino, L., 2016. Effects of multi-stressors on juveniles of the marine fish *Pomatoschistus microps*: gold nanoparticles, microplastics and temperature. *Aquat. Toxicol.* 170:89–103. <https://doi.org/10.1016/j.aquatox.2015.11.011>.
- Fleiss, J.L., 1981. *Statistical Methods for Rates and Proportions*. 2nd ed. Wiley, New York.
- Foekema, E.M., De Groot, C., Mergia, M.T., van Franeker, J.A., Murk, A.J., Koelmans, A.A., 2013. Plastic in north sea fish. *Environ. Sci. Technol.* 47:8818–8824. <https://doi.org/10.1021/es400931b>.
- Fonte, E., Ferreira, P., Guilhermino, L., 2016. Temperature rise and microplastics interact with the toxicity of the antibiotic cefalexin to juveniles of the common goby (*Pomatoschistus microps*): post-exposure predatory behaviour, acetylcholinesterase activity and lipid peroxidation. *Aquat. Toxicol.* 180:173–185. <https://doi.org/10.1016/j.aquatox.2016.09.015>.
- Free, C.M., Jensen, O.P., Mason, S.A., Eriksen, M., Williamson, N.J., Boldgiv, B., 2014. High-levels of microplastic pollution in a large, remote, mountain lake. *Mar. Pollut. Bull.* 85:156–163. <https://doi.org/10.1016/j.marpolbul.2014.06.001>.
- Graham, E.R., Thompson, J.T., 2009. Deposit- and suspension-feeding sea cucumbers (Echinodermata) ingest plastic fragments. *J. Exp. Mar. Biol. Ecol.* 368:22–29. <https://doi.org/10.1016/j.jembe.2008.09.007>.
- Green, D.S., 2016. Effects of microplastics on European flat oysters, *Ostrea edulis* and their associated benthic communities. *Environ. Pollut.* 216:95–103. <https://doi.org/10.1016/j.envpol.2016.05.043>.
- Green, D.S., Boots, B., Sigwart, J., Jiang, S., Rocha, C., 2016. Effects of conventional and biodegradable microplastics on a marine ecosystem engineer (*Arenicola marina*) and sediment nutrient cycling. *Environ. Pollut.* 208:426–434. <https://doi.org/10.1016/j.envpol.2015.10.010>.
- Gregory, M.R., 1978. Accumulation and distribution of virgin plastic granules on New Zealand beaches. *N. Z. J. Mar. Freshw. Res.* 12:399–414. <https://doi.org/10.1080/00288330.1978.9515768>.
- Gurevitch, J., Curtis, P.S., Jones, M.H., 2001. Meta-analysis in ecology. *Advances in Ecological Research*. Academic Press:pp. 199–247. [https://doi.org/10.1016/S0065-2504\(01\)32013-5](https://doi.org/10.1016/S0065-2504(01)32013-5).
- Gutow, L., Eckerlebe, A., Giménez, L., Saborowski, R., 2016. Experimental evaluation of seaweeds as a vector for microplastics into marine food webs. *Environ. Sci. Technol.* 50:915–923. <https://doi.org/10.1021/acs.est.5b02431>.
- Hämer, J., Gutow, L., Köhler, A., Saborowski, R., 2014. Fate of microplastics in the marine isopod *Idotea emarginata*. *Environ. Sci. Technol.* 48:13451–13458. <https://doi.org/10.1021/es501385y>.
- Hedges, L.V., Olkin, I., 1985. *Statistical methods for meta-analysis*. Academic Press, San Diego, CA.
- Imhof, H.K., Laforst, C., 2016. Hazardous or not - are adult and juvenile individuals of *Potamopyrgus antipodarum* affected by non-buoyant microplastic particles? *Environ. Pollut.* 218:383–391. <https://doi.org/10.1016/j.envpol.2016.07.017>.
- Jemec, A., Horvat, P., Kunej, U., Bele, M., Kržan, A., 2016. Uptake and effects of microplastic textile fibers on freshwater crustacean *Daphnia magna*. *Environ. Pollut.* 219:201–209. <https://doi.org/10.1016/j.envpol.2016.10.037>.
- Jeong, C.-B., Won, E.-J., Kang, H.-M., Lee, M.-C., Hwang, D.-S., Hwang, U.-K., Zhou, B., Souissi, S., Lee, S.-J., Lee, J.-S., 2016. Microplastic size-dependent toxicity, oxidative stress induction, and p-JNK and p-p38 activation in the monogonot rotifer (*Brachionus koreanus*). *Environ. Sci. Technol.* 50:8849–8857. <https://doi.org/10.1021/acs.est.6b01441>.
- Kaposi, K.L., Mos, B., Kelaher, B.P., Dworjanyn, S.A., 2014. Ingestion of microplastic has limited impact on a marine larva. *Environ. Sci. Technol.* 48:1638–1645. <https://doi.org/10.1021/es404295e>.
- Karami, A., Romano, N., Galloway, T., Hamzah, H., 2016. Virgin microplastics cause toxicity and modulate the impacts of phenanthrene on biomarker responses in African catfish (*Clarias gariepinus*). *Environ. Res.* 151:58–70. <https://doi.org/10.1016/j.envres.2016.07.024>.
- Kashiwada, S., 2006. Distribution of nanoparticles in the see-through Medaka (*Oryzias latipes*). *Environ. Health Perspect.* 114:1697–1702. <https://doi.org/10.1289/ehp.9209>.
- Kershaw, P.J., Rochman, C.M., 2015. *Sources, fate and effects of microplastics in the marine environment: part 2 of a global assessment. Reports and Studies - IMO/FAO/Unesco-IOC/WMO/IAEA/UN/UNEP Joint Group of Experts on the Scientific Aspects of Marine Environmental Protection (GESAMP)* eng no. 93.
- Khan, F.R., Syberg, K., Shashoua, Y., Bury, N.R., 2015. Influence of polyethylene microplastic beads on the uptake and localization of silver in zebrafish (*Danio rerio*). *Environ. Pollut.* 206:73–79. <https://doi.org/10.1016/j.envpol.2015.06.009>.
- Koelmans, A.A., Bakir, A., Burton, G.A., Janssen, C.R., 2016. Microplastic as a vector for chemicals in the aquatic environment: critical review and model-supported reinterpretation of empirical studies. *Environ. Sci. Technol.* 50:3315–3326. <https://doi.org/10.1021/acs.est.5b06069>.
- Laist, D.W., 1997. Impacts of marine debris: entanglement of marine life in marine debris including a comprehensive list of species with entanglement and ingestion records. *Marine Debris, Springer Series on Environmental Management*. Springer, New York, NY:pp. 99–139. https://doi.org/10.1007/978-1-4613-8486-1_10.
- Lu, Y., Zhang, Y., Deng, Y., Jiang, W., Zhao, Y., Geng, J., Ding, L., Ren, H., 2016. Uptake and accumulation of polystyrene microplastics in zebrafish (*Danio rerio*) and toxic effects in liver. *Environ. Sci. Technol.* 50:4054–4060. <https://doi.org/10.1021/acs.est.6b00183>.
- Luís, L.G., Ferreira, P., Fonte, E., Oliveira, M., Guilhermino, L., 2015. Does the presence of microplastics influence the acute toxicity of chromium(VI) to early juveniles of the common goby (*Pomatoschistus microps*)? A study with juveniles from two wild estuarine populations. *Aquat. Toxicol.* 164:163–174. <https://doi.org/10.1016/j.aquatox.2015.04.018>.
- Lusher, A.L., Welden, N.A., Sobral, P., Cole, M., 2017. Sampling, isolating and identifying microplastics ingested by fish and invertebrates. *Anal. Methods* 9:1346–1360. <https://doi.org/10.1039/C6AY02415G>.
- Ma, Y., Huang, A., Cao, S., Sun, F., Wang, L., Guo, H., Ji, R., 2016. Effects of nanoplastics and microplastics on toxicity, bioaccumulation, and environmental fate of phenanthrene in fresh water. *Environ. Pollut.* 219:166–173. <https://doi.org/10.1016/j.envpol.2016.10.061>.
- Mazurais, D., Ernande, B., Quazuguel, P., Severe, A., Huelvan, C., Madec, L., Mouchel, O., Soudant, P., Robbens, J., Huvet, A., Zambonino-Infante, J., 2015. Evaluation of the impact of polyethylene microbeads ingestion in European sea bass (*Dicentrarchus labrax*) larvae. *Mar. Environ. Res.* 112:78–85. <https://doi.org/10.1016/j.marenvres.2015.09.009>.
- Murray, F., Cowie, P.R., 2011. Plastic contamination in the decapod crustacean *Nephrops norvegicus* (Linnaeus, 1758). *Mar. Pollut. Bull.* 62:1207–1217. <https://doi.org/10.1016/j.marpolbul.2011.03.032>.
- Nasser, F., Lynch, I., 2016. Secreted protein eco-corona mediates uptake and impacts of polystyrene nanoparticles on *Daphnia magna*. *J. Proteome* 137:45–51. <https://doi.org/10.1016/j.jprot.2015.09.005>.
- Nobre, C.R., Santana, M.F.M., Maluf, A., Cortez, F.S., Cesar, A., Pereira, C.D.S., Turra, A., 2015. Assessment of microplastic toxicity to embryonic development of the sea urchin *Lytechinus variegatus* (Echinodermata: Echinoidea). *Mar. Pollut. Bull.* 92:99–104. <https://doi.org/10.1016/j.marpolbul.2014.12.050>.
- Oberbeckmann, S., Löder, M.G.J., Labrenz, M., 2015. Marine microplastic-associated biofilms – a review. *Environ. Chem.* 12:551–562. <https://doi.org/10.1071/EN15069>.
- Ogonowski, M., Schür, C., Jarsén, A., Gorokhova, E., 2016. The effects of natural and anthropogenic microparticles on individual fitness in *Daphnia magna*. *PLoS ONE* 11, e0155063. <https://doi.org/10.1371/journal.pone.0155063>.
- Oliveira, M., Ribeiro, A., Hylland, K., Guilhermino, L., 2013. Single and combined effects of microplastics and pyrene on juveniles (0+ group) of the common goby *Pomatoschistus microps* (Teleostei, Gobiidae). *Ecol. Indic.* 34:641–647. <https://doi.org/10.1016/j.ecolind.2013.06.019>.
- Pace, M.L., Cole, J.J., Carpenter, S.R., Kitchell, J.F., 1999. Trophic cascades revealed in diverse ecosystems. *Trends Ecol. Evol.* 14:483–488. [https://doi.org/10.1016/S0169-5347\(99\)01723-1](https://doi.org/10.1016/S0169-5347(99)01723-1).
- Paine, R.T., 1992. Food-web analysis through field measurement of per capita interaction strength. *Nature* 355:73. <https://doi.org/10.1038/355073a0>.
- Pallone, F., 2015. Text - H.R.1321 - 114th Congress (2015–2016): Microbead-Free Waters Act of 2015. [WWW Document]. URL: <https://www.congress.gov/bills/114th-congress/house-bill/1321/text>, Accessed date: 5 December 2017.
- Paul-Pont, I., Lacroix, C., González Fernández, C., Hégaret, H., Lambert, C., Le Goïc, N., Frère, L., Cassone, A.-L., Sussarellu, R., Fabioux, C., Guyomarch, J., Albentosa, M., Huvet, A., Soudant, P., 2016. Exposure of marine mussels *Mytilus* spp. to polystyrene microplastics: toxicity and influence on fluoranthene bioaccumulation. *Environ. Pollut.* 216:724–737. <https://doi.org/10.1016/j.envpol.2016.06.039>.
- Pedà, C., Caccamo, L., Fossi, M.C., Gai, F., Andaloro, F., Genovesi, L., Perdichizzi, A., Romeo, T., Maricchiolo, G., 2016. Intestinal alterations in European sea bass *Dicentrarchus labrax* (Linnaeus, 1758) exposed to microplastics: preliminary results. *Environ. Pollut.* 212:251–256. <https://doi.org/10.1016/j.envpol.2016.01.083>.
- Phillips, M.B., Bonner, T.H., 2015. Occurrence and amount of microplastic ingested by fishes in watersheds of the Gulf of Mexico. *Mar. Pollut. Bull.* 100:264–269. <https://doi.org/10.1016/j.marpolbul.2015.08.041>.
- Phuong, N.N., Zalouk-Vernoux, A., Poirier, L., Kamari, A., Châtel, A., Mouneyrac, C., Lagarde, F., 2016. Is there any consistency between the microplastics found in the field and those used in laboratory experiments? *Environ. Pollut.* 211:111–123. <https://doi.org/10.1016/j.envpol.2015.12.035>.

- Powell, M.D., Berry, A.J., 1990. Ingestion and regurgitation of living and inert materials by the estuarine copepod *Eurytemora affinis* (Poppe) and the influence of salinity. *Estuar. Coast. Shelf Sci.* 31:763–773. [https://doi.org/10.1016/0272-7714\(90\)90081-2](https://doi.org/10.1016/0272-7714(90)90081-2).
- R Core Team, 2016. R: A Language and Environment for Statistical Computing. R Foundation for Statistical Computing, Vienna, Austria URL. <https://www.R-project.org>.
- Rehse, S., Kloas, W., Zarfl, C., 2016. Short-term exposure with high concentrations of pristine microplastic particles leads to immobilisation of *Daphnia magna*. *Chemosphere* 153:91–99. <https://doi.org/10.1016/j.chemosphere.2016.02.133>.
- Rist, S.E., Assidqi, K., Zamani, N.P., Appel, D., Perschke, M., Huhn, M., Lenz, M., 2016. Suspended micro-sized PVC particles impair the performance and decrease survival in the Asian green mussel *Perna viridis*. *Mar. Pollut. Bull.* 111:213–220. <https://doi.org/10.1016/j.marpolbul.2016.07.006>.
- Rochman, C.M., Hoh, E., Kurobe, T., Teh, S.J., 2013. Ingested plastic transfers hazardous chemicals to fish and induces hepatic stress. *Sci. Rep.* 3 (3263). <https://doi.org/10.1038/srep03263>.
- Rochman, C.M., Regan, F., Thompson, R.C., 2017. On the harmonization of methods for measuring the occurrence, fate and effects of microplastics. *Anal. Methods* 9: 1324–1325. <https://doi.org/10.1039/C7AY90014G>.
- Rummel, C.D., Adolfsson-Erici, M., Jahnke, A., MacLeod, M., 2016. No measurable “cleaning” of polychlorinated biphenyls from Rainbow Trout in a 9 week depuration study with dietary exposure to 40% polyethylene microspheres. *Environ. Sci.: Processes Impacts* 18:788–795. <https://doi.org/10.1039/C6EM00234J>.
- Schneider, C.A., Rasband, W.S., Eliceiri, K.W., 2012. NIH Image to ImageJ: 25 years of image analysis. *Nat. Methods* 9 (7), 671–675 (PMID 22930834 (on Google Scholar)).
- Seltenrich, N., 2015. New link in the food chain? Marine plastic pollution and seafood safety. *Environ. Health Perspect.* 123:A34–A41. <https://doi.org/10.1289/ehp.123-A34>.
- Setälä, O., Fleming-Lehtinen, V., Lehtiniemi, M., 2014. Ingestion and transfer of microplastics in the planktonic food web. *Environ. Pollut.* 185:77–83. <https://doi.org/10.1016/j.envpol.2013.10.013>.
- Sheldon, R.W., Sutcliffe Jr., W.H., Paranjape, M.A., 1977. Structure of pelagic food chain and relationship between plankton and fish production. *J. Fish. Res. Board Can.* 34: 2344–2353. <https://doi.org/10.1139/f77-314>.
- Snell, T.W., Hicks, D.G., 2011. Assessing toxicity of nanoparticles using *Brachionus manjavacas* (Rotifera). *Environ. Toxicol.* 26:146–152. <https://doi.org/10.1002/tox.20538>.
- Sogard, S.M., 1997. Size-selective mortality in the juvenile stage of teleost fishes: a review. *Bull. Mar. Sci.* 60, 1129–1157.
- Sommer, U., Stibor, H., Katechakis, A., Sommer, F., Hansen, T., 2002. Pelagic food web configurations at different levels of nutrient richness and their implications for the ratio fish production:primary production. *Hydrobiologia* 484:11–20. <https://doi.org/10.1023/A:1021340601986>.
- Sussarellu, R., Suquet, M., Thomas, Y., Lambert, C., Fabioux, C., Pernet, M.E.J., Goïc, N.L., Quillien, V., Mingant, C., Epelboin, Y., Corporeau, C., Guyomarch, J., Robbins, J., Paul Pont, I., Soudant, P., Huvet, A., 2016. Oyster reproduction is affected by exposure to polystyrene microplastics. *PNAS* 113:2430–2435. <https://doi.org/10.1073/pnas.1519019113>.
- Tosetto, L., Brown, C., Williamson, J.E., 2016. Microplastics on beaches: ingestion and behavioural consequences for beachhoppers. *Mar. Biol.* 163 (199). <https://doi.org/10.1007/s00227-016-2973-0>.
- Ugolini, A., Ungherese, G., Ciofini, M., Lapucci, A., Camaiti, M., 2013. Microplastic debris in sandhoppers. *Estuar. Coast. Shelf Sci.* 129:19–22. <https://doi.org/10.1016/j.ecss.2013.05.026>.
- Van Cauwenberghe, L., Janssen, C.R., 2014. Microplastics in bivalves cultured for human consumption. *Environ. Pollut.* 193:65–70. <https://doi.org/10.1016/j.envpol.2014.06.010>.
- Van Cauwenberghe, L., Claessens, M., Vandegehuchte, M.B., Janssen, C.R., 2015. Microplastics are taken up by mussels (*Mytilus edulis*) and lugworms (*Arenicola marina*) living in natural habitats. *Environ. Pollut.* 199:10–17. <https://doi.org/10.1016/j.envpol.2015.01.008>.
- Viechtbauer, W., 2010. Conducting Meta-Analyses in R with the metafor Package. *J. Stat. Softw.* 3:1–48. [10.18637/jss.v036.i03](https://doi.org/10.18637/jss.v036.i03).
- Wardrop, P., Shimeta, J., Nuggeoda, D., Morrison, P.D., Miranda, A., Tang, M., Clarke, B.O., 2016. Chemical pollutants sorbed to ingested microbeads from personal care products accumulate in fish. *Environ. Sci. Technol.* 50:4037–4044. <https://doi.org/10.1021/acs.est.5b06280>.
- Watts, A.J.R., Lewis, C., Goodhead, R.M., Beckett, S.J., Moger, J., Tyler, C.R., Galloway, T.S., 2014. Uptake and retention of microplastics by the shore crab *Carcinus maenas*. *Environ. Sci. Technol.* 48:8823–8830. <https://doi.org/10.1021/es501090e>.
- Watts, A.J.R., Urbina, M.A., Goodhead, R., Moger, J., Lewis, C., Galloway, T.S., 2016. Effect of microplastic on the gills of the shore crab *Carcinus maenas*. *Environ. Sci. Technol.* 50: 5364–5369. <https://doi.org/10.1021/acs.est.6b01187>.
- Wegner, A., Besseling, E., Foekema, E.M., Kamermans, P., Koelmans, A.a., 2012. Effects of nanoplastyrene on the feeding behavior of the blue mussel (*Mytilus edulis* L.). *Environ. Toxicol. Chem.* 31:2490–2497. <https://doi.org/10.1002/etc.1984>.
- Welden, N.A.C., Cowie, P.R., 2016. Long-term microplastic retention causes reduced body condition in the langoustine, *Nephrops norvegicus*. *Environ. Pollut.* 218:895–900. <https://doi.org/10.1016/j.envpol.2016.08.020>.
- Wright, S.L., Rowe, D., Thompson, R.C., Galloway, T.S., 2013a. Microplastic ingestion decreases energy reserves in marine worms. *Curr. Biol.* 23:R1031–R1033. <https://doi.org/10.1016/j.cub.2013.10.068>.
- Wright, S.L., Thompson, R.C., Galloway, T.S., 2013b. The physical impacts of microplastics on marine organisms: a review. *Environ. Pollut.* 178:483–492. <https://doi.org/10.1016/j.envpol.2013.02.031>.